

V4.2 Toy Model 4A

From Relational Distance Matrix to Emergent Metric Tensor Candidate

Rev A - Exploratory Gravity / Metric Bridge Report

Date: 2026.05.26 | Continuation after Toy Model 3G

Status: Toy Model 4A tests whether local metric components can be extracted from relational graph distances. It also records a critical failure/guardrail: a positive shortest-path distance matrix alone produces a Riemannian spatial metric, not a Lorentzian spacetime metric.

1. Purpose

Toy Model 1D showed that a relational graph can generate valid shortest-path distances and operational lensing analogues. Toy Model 4A now asks the next gravity-side question: can local metric-tensor components be extracted from relational distances?

The target is not yet General Relativity. The target is the first mathematical bridge from distance structure to a local metric candidate.

2. Term reminders

- Distance matrix: a table of shortest-path distances between graph nodes.
- Metric tensor, written $g_{\mu\nu}$: the local rule that tells geometry how to measure squared interval from small coordinate changes.
- Riemannian metric: positive-distance geometry, like ordinary spatial geometry.
- Lorentzian metric: spacetime geometry with one time-like sign and three space-like signs, usually written $(-+++)$.
- Signature: the pattern of positive and negative eigenvalues of the metric tensor.
- Smoothness limit: the dense-network limit where local discrete data approximates a continuous manifold.

3. Test graph

A 2D grid graph was used as a controlled test. Each node has coordinates (x,y) . Edge costs were assigned from a known conformal line element:

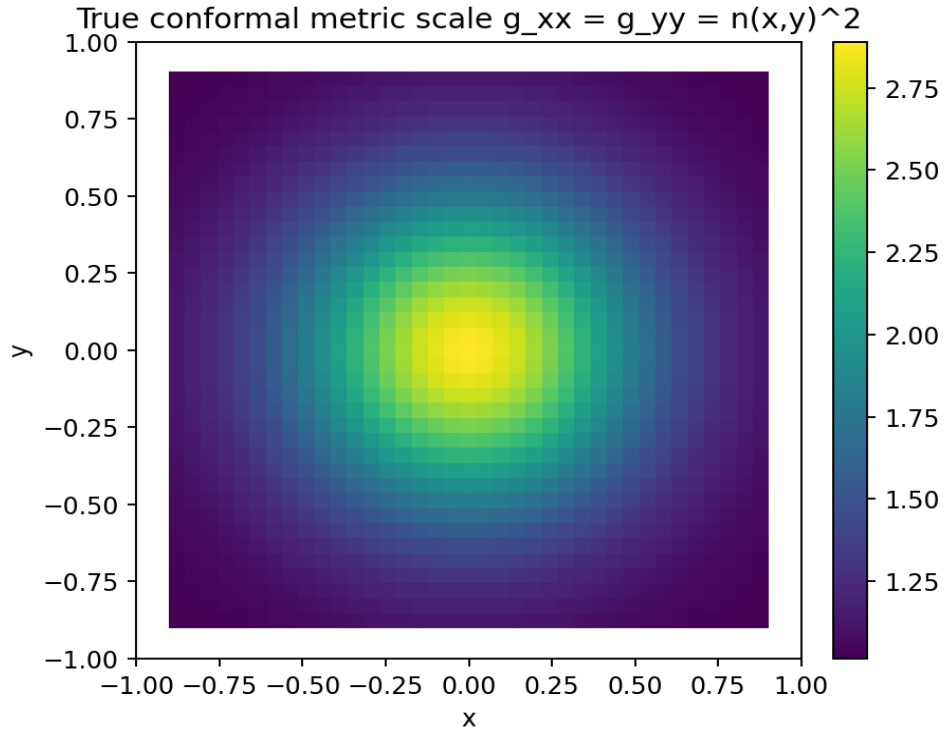
$$ds = n(x,y) \sqrt{dx^2 + dy^2}$$

$$n(x,y) = 1 + A \exp[-(x^2 + y^2)/(2 \rho^2)]$$

Therefore:

$$g_{ab}(x,y) = n(x,y)^2 \delta_{ab}$$

This gives a known ground truth metric for checking the extraction. The point is not that the universe has this metric. The point is to verify whether the extraction operator can recover a local metric from graph distances.



4. Metric extraction operator

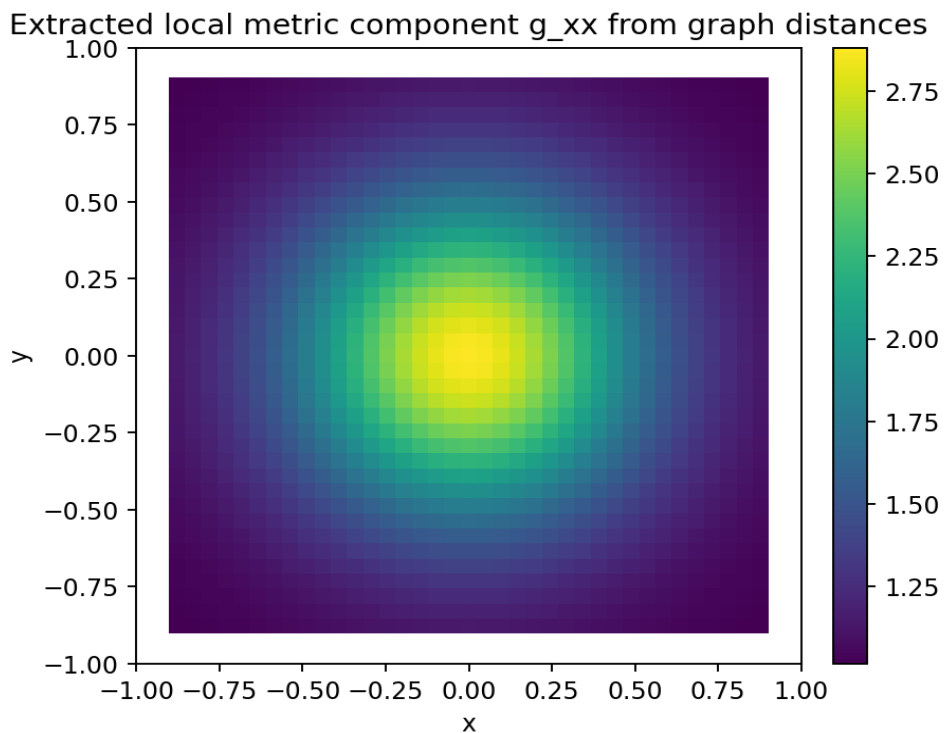
For a node i and nearby node j , the local squared distance is approximated by:

$$d_{ij}^2 \approx g_{xx} dx^2 + 2 g_{xy} dx dy + g_{yy} dy^2$$

Meaning of symbols:

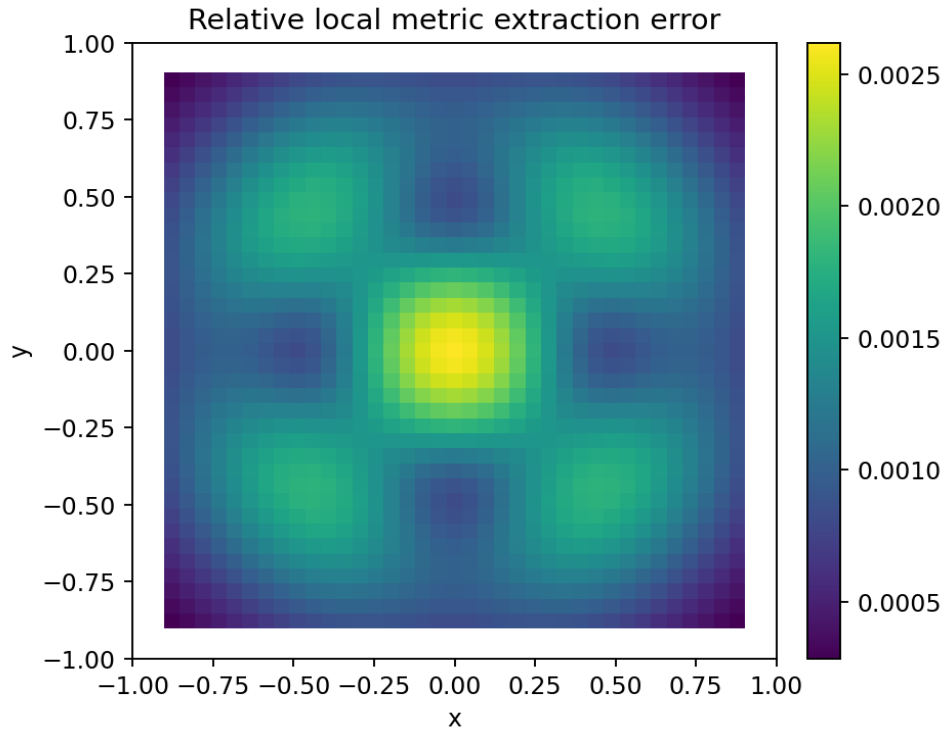
- d_{ij} is the relational shortest-path distance between node i and node j .
- dx and dy are small coordinate differences between the two nodes.
- g_{xx} , g_{xy} , and g_{yy} are the local metric components to be extracted.
- The factor 2 in $2 g_{xy} dx dy$ appears because the symmetric off-diagonal terms g_{xy} and g_{yx} contribute together.

Using several nearby distances, the unknown metric components are solved by least squares. In higher dimensions, the same idea fits all independent components of $g_{\mu\nu}$.



5. Extraction accuracy

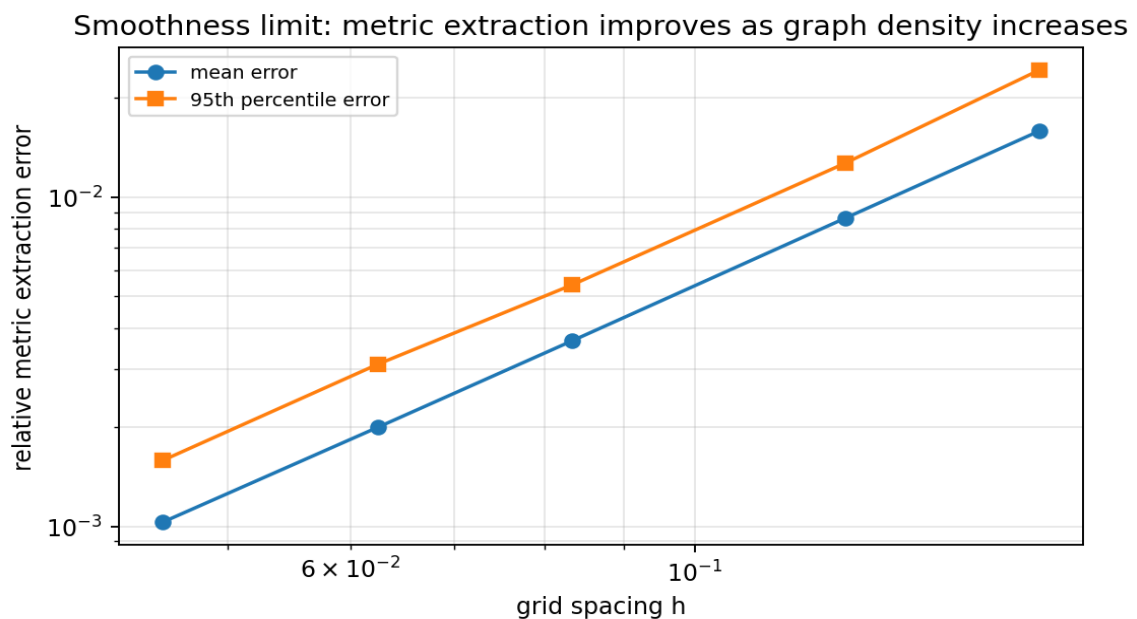
The extracted g_{xx} field matches the controlled ground-truth metric well. The map below shows relative local error.



Diagnostic	Value
Main grid size	41 x 41
Interior fitted metric points	1369
Mean relative metric error	1.255307e-03
95th percentile relative error	1.946092e-03
Max relative error	2.619859e-03
Mean $ g_{xy} $ extracted	1.196268e-03
Max $ g_{xy} $ extracted	2.913362e-03

6. Smoothness limit

The graph was refined from coarse to dense grids. As grid spacing h decreases, the extracted local metric better approximates the continuous metric. This is the required smoothness behavior for a metric candidate.

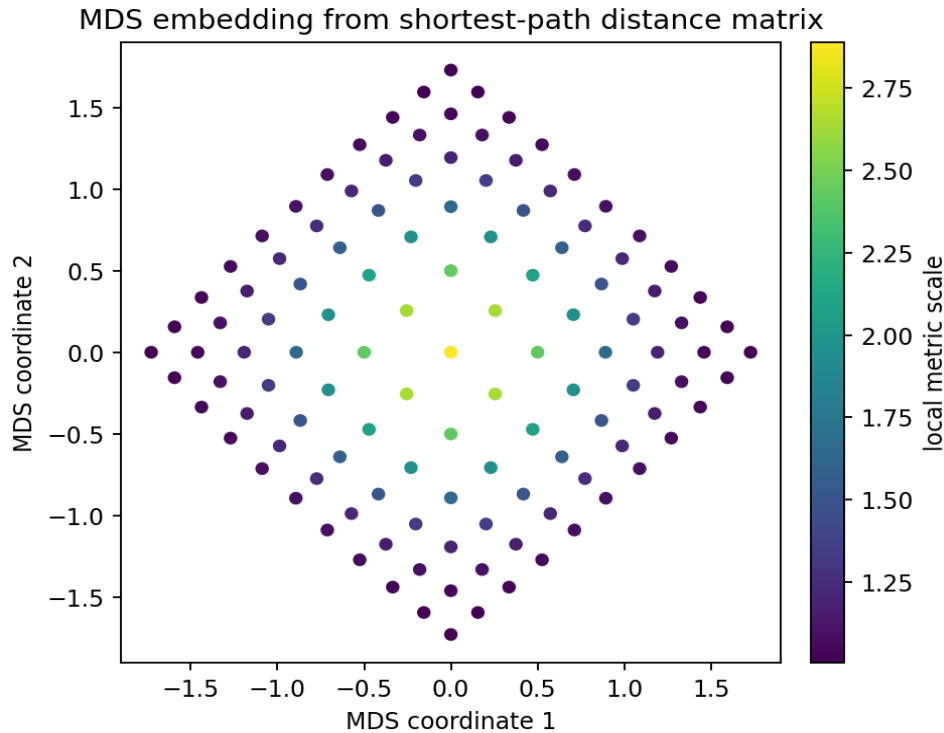


N	h	mean err	95% err	max err	min eigenvalue	max anisotropy
13	0.16667	1.586781e-02	2.430128e-02	2.842774e-02	1.110479e+00	1.811162e-02
17	0.12500	8.627043e-03	1.267485e-02	1.617324e-02	1.056937e+00	1.045426e-02
25	0.08333	3.654732e-03	5.405107e-03	7.246963e-03	1.027149e+00	4.814752e-03
33	0.06250	1.997918e-03	3.105484e-03	4.088103e-03	1.018193e+00	2.726773e-03
45	0.04545	1.030276e-03	1.586496e-03	2.166059e-03	1.012914e+00	1.439102e-03

7. Distance matrix and MDS embedding

The global shortest-path distance matrix was sampled and embedded into 2D using classical multidimensional scaling (MDS).

Term reminder: MDS means a method that assigns coordinates to points so that Euclidean distances between assigned coordinates approximate the input distance matrix.

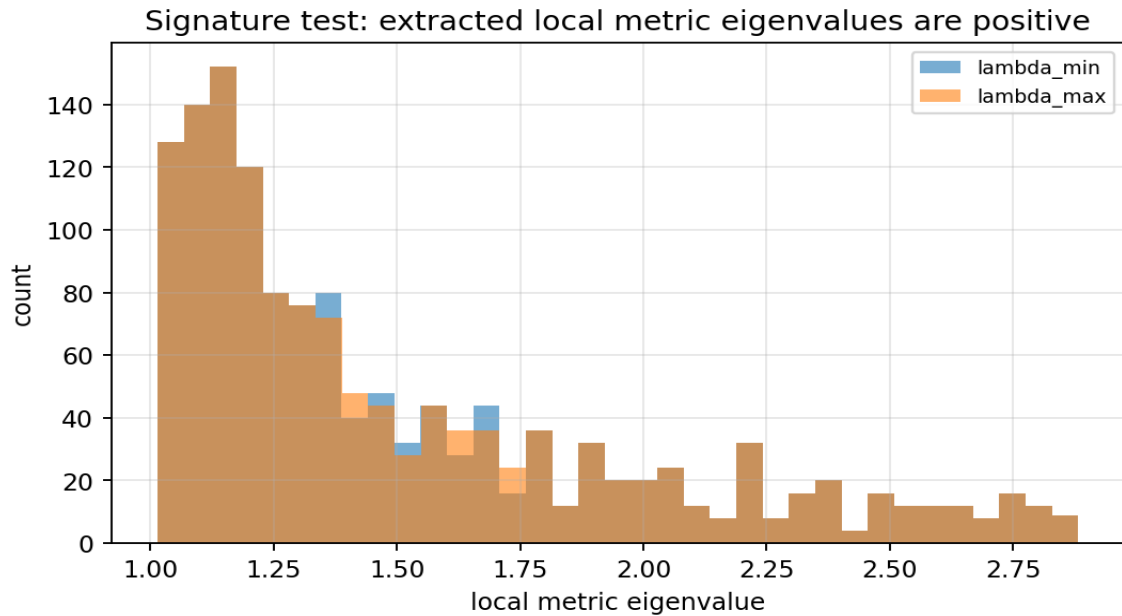


Embedding diagnostic	Value
Sampled nodes	121
MDS stress, flat graph	2.780269e-02
MDS stress, conformal source graph	8.410125e-02
Positive MDS eigenvalues above threshold	51
Top 6 MDS eigenvalues	8.5899e+01, 8.5899e+01, 6.5015e+00, 5.1977e+00, 5.1977e+00, 1.5761e+00

Interpretation: the global graph distance structure can be embedded approximately, but the source-weighted graph is less globally flat than the baseline graph. This supports the idea that local metric extraction is more reliable than forcing one global flat embedding.

8. Signature test

This is the key guardrail of Toy Model 4A. The extracted local metric eigenvalues are positive. That means the distance graph produced a Riemannian spatial metric candidate, not a Lorentzian spacetime metric.



Signature diagnostic	Value
Positive eigenvalue count	2738
Zero-ish eigenvalue count	0
Negative eigenvalue count	0
Minimum eigenvalue	1.014169e+00
Maximum eigenvalue	2.882429e+00

Why this matters: an ordinary shortest-path distance matrix is non-negative and obeys triangle inequality. That structure naturally produces positive local distance geometry. It does not by itself create the negative time-like direction required by relativity.

9. Result inside the V4.2 framework

Toy Model 4A supports this limited ladder:

```
relational adjacency
-> edge costs
-> shortest-path distance matrix
-> local quadratic distance fit
-> Riemannian metric candidate g_ab
-> smoothness test
-> future curvature candidate
```

But Toy Model 4A does not yet support this stronger ladder:

```
relational adjacency alone
-> Lorentzian spacetime metric
-> Einstein curvature
-> gravity
```

To get Lorentzian signature, the next model needs additional causal/time structure: directed relations, event ordering, proper-time-like intervals, or a split between spatial distance and temporal update direction.

10. What Toy Model 4A achieves

- It defines an explicit operator for extracting local metric components from relational distances.
- It verifies the operator on a controlled graph with known ground-truth metric.
- It shows the extracted metric improves in the dense-network smoothness limit.
- It confirms positive-definite Riemannian signature from ordinary graph distances.
- It identifies the exact missing ingredient for relativity: causal or signed interval structure.

11. What Toy Model 4A does not achieve

- It does not derive a Lorentzian metric.
- It does not derive Einstein Field Equations.
- It does not derive curvature dynamics.

- It does not explain gravity yet.
- It does not connect to dark matter or dark energy yet.
- It does not prove spacetime is a graph.
- It does not prove physical 3T existence.

12. Close-out statement for 4A

Toy Model 4A successfully demonstrates that a local Riemannian metric candidate can be extracted from relational graph distances by fitting local squared distances to a quadratic form. The extraction converges as graph density increases and passes positive-definiteness checks. However, the signature test is decisive: a positive shortest-path distance matrix alone does not produce Lorentzian spacetime signature. Toy Model 4A therefore bridges relational distance to spatial metric candidate, but it also shows that the next step must introduce causal or signed interval structure before General Relativity can be approached.

13. Recommended next chapter

The next logical chapter is Toy Model 4B: Causal Relational Graph and Lorentzian Signature Candidate.

1. Add event ordering or directed causal relations to the relational graph.
2. Separate spatial distance from temporal update/proper-time structure.
3. Construct a signed interval candidate $ds^2 = -c^2 dt^2 + g_{ab} dx^a dx^b$.
4. Test whether the local fitted metric can show one negative and three positive directions.
5. Only after Lorentzian signature exists should curvature and Einstein-like equations be tested.

14. Rebind prompt for continuation

We are resuming after Toy Model 4A.

4A tested whether a local metric tensor candidate can be extracted from a relational shortest-path distance matrix.

Local extraction rule:
 $d_{ij}^2 \approx g_{xx} dx^2 + 2 g_{xy} dx dy + g_{yy} dy^2$

Result:
 A local Riemannian metric candidate can be extracted from graph distances.
 The extraction improves as graph density increases.

Critical guardrail:
 A positive shortest-path distance matrix alone produces positive-definite Riemannian geometry. It does not produce Lorentzian spacetime signature (-+++).

Therefore:
 relational distances -> spatial metric candidate
 but not yet:
 relational distances -> spacetime metric -> GR

Next recommended chapter:
 Toy Model 4B - Causal Relational Graph and Lorentzian Signature Candidate.

Main 4B target:
 Add causal/time structure so that a signed interval can be tested:
 $ds^2 = -c^2 dt^2 + g_{ab} dx^a dx^b$